

Ho Ko

AI / ML SOFTWARE ENGINEER

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AI/ML Software Engineer with cross-domain experience spanning astronomy, weather forecasting, economics, healthcare, NLP, and computer vision. Works across the full **model lifecycle** — training and distilling deep models, architecting highly efficient inference engines, and building scalable, high-throughput serving — and brings that intelligence **on-device** through WebGPU, Three.js, and local LLMs in privacy-preserving systems that have reached **1B+ users**.

EXPERIENCE

- Machine Learning Engineer · Google** 2026–Present
- Build **knowledge-grounded retrieval & ranking models** on Google Lens that power AI Mode in Google Search and AI Overviews at Search scale.
 - Distill Gemini** into highly compressed models that hold near-equal quality while meeting the Lens stack's latency, scalability, and multimodal requirements.
- Software Engineer · Google** 2023–2026
- Core contributor to the **LiteRT LM inference engine**, running LLMs privately and offline across Windows, macOS, Linux, and Android — including NPU acceleration.
 - Enabled Gemma multimodality (vision + audio + text) and Gemini Nano on-device; helped ship **Chrome Built-in AI and Google AI Core, reaching 1B+ users**.
 - Shipped the **Google AI Edge Gallery** app on Android and iOS.
- Research Collaborator · Google DeepMind** 2024–2025
- Designed **Sparse RAG (ICLR 2025)**, a sparse decoding paradigm that encodes retrieved docs in parallel and attends only to the most relevant caches via learned control tokens.
 - Cut long-range attention latency while **improving generation quality across 4 retrieval benchmarks**.
- Research Collaborator · Google Health & Fitbit** 2024–2025
- Built and evaluated ML models **detecting hypertension risk from raw wrist PPG** passively collected on a consumer smartwatch (medRxiv), enabling non-invasive screening at population scale.
- Software Engineer · MyCareLinq** 2023
- Built **NLP pipelines and a home-health-care knowledge graph** from unstructured health documentation.
 - Integrated LLMs powering caregiving assistants that surface contextual recommendations and care-plan summaries.
- Software Engineer Intern · Google** 2022
- Built a TFLite tensor extractor + metadata-writer with classification rules at **100% coverage of common GenAI task libraries** (LiteRT Task Library).
 - Integrated a **C++ code generator** compiling optimized inference pipelines for arbitrary TFLite files.
- Research Assistant · Caltech (SURF)** 2018
- Analyzed transient imaging from the Zwicky Transient Facility; **co-authored a Type IIb supernova study in ApJL (878:L5)** and simulated double-peak light-curve models.

EDUCATION

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| M.S. Computer Science
USC — AI & ML · GPA 4.0/4.0
2021–2023 | M.S. Electrical & Computer Eng.
Waterloo — ML & AI · GPA 3.9/4.0
2020–2021 |
| B.S. Computer Science
NCU, Taiwan · GPA 3.8/4.0
2016–2020 | |

SELECTED PUBLICATIONS

- Accelerating Inference of Retrieval-Augmented Generation via Sparse Context Selection* — **ICLR 2025**, Google DeepMind.
- Opportunistic Hypertension Detection from Wearable PPG* — **medRxiv 2025**, Google Health & Fitbit.
- A Type IIb Supernova (ZTF18aalrxas)* — **ApJL 878:L5**, 2019.
- Real-world WebAssembly Performance for ML* — **ACM IMC 2023**.

TECHNICAL SKILLS

- Languages:** Python, C/C++, Swift, Java, Kotlin, JavaScript, x86 Assembly, MATLAB
- ML Lifecycle:** Model Training, Distillation, Quantization & Compression, Inference Engines, On-Device Deployment, Scalable Serving
- On-Device & Web:** WebGPU, Three.js, Transformers.js, MediaPipe, WebRTC, Local LLMs (Gemma, LFM)
- Domains:** NLP, Computer Vision, Healthcare, Astronomy, Weather Forecasting, Economics